

Unit 3 Problem Solving and DA Practice Test Module II

1

★ Mark for Review

ABC

The number p is 60% of the positive number q , and p is 390% of the number r . If q is $x\%$ of r , what is the value of x ?

2

★ Mark for Review

ABC

At an auction, the starting price of a gemstone is set at 60% less than its assessed value of \$30,000. Eli makes an offer that is 370% of the starting price. What is the value of Eli's offer, in dollars?

3

★ Mark for Review

ABC

The number r is 70% less than the number s . The number r is 60% greater than the number t . If $s = 180$, what is the value of t ?

4

★ Mark for Review

ABC

The number x is 75% less than the positive number y . The number z is 84% greater than x . The number z is how many times y ?

5

★ Mark for Review

ABC

In 2010, the number of houses built in Town A was n percent greater than the number of houses built in Town B. If 700 houses were built in Town A in 2010, which expression represents the number of houses built in Town B in 2010?

A. $700 \left(\frac{100 + n}{100} \right)$

B. $700 \left(\frac{100}{100 + n} \right)$

C. $700 \left(\frac{100 - n}{100} \right)$

D. $700 \left(\frac{100}{100 - n} \right)$

6

★ Mark for Review

ABC

The area of Wompatuck State Park in Massachusetts is 5.5 square miles. Which of the following is closest to the area of this park, in square kilometers? (Use 1 kilometer = 0.62 mile.)

A. 2.1

B. 3.4

C. 8.9

D. 14.3

7

★ Mark for Review

ABC

If a apples cost d dollars, which of the following expressions gives the cost of 20 apples, in dollars?

A. $\frac{20a}{d}$

B. $\frac{20d}{a}$

C. $20d$

D. $\frac{20}{ad}$

8

★ Mark for Review

ABC

A model of a building has a scale where a width of 5 inches in the model represents an actual width of 16 feet. A room in the building has an actual width of $10w$ feet. Which expression represents the width, in inches, of the room in the model?

A. $\frac{5}{8}w$

B. $\frac{25}{8}w$

C. $8w$

D. $32w$

9

★ Mark for Review

ABC

An object is expanding so that its surface area is increasing at a constant rate of 3 square inches per hour. There are 144 square inches in 1 square foot. Of the following, which is closest to the rate at which the surface area of the object is increasing, in square meters per hour? (Use 1 foot = 0.3048 meter.)

A. 0.0019

B. 0.0064

C. 0.0684

D. 0.2242

10

★ Mark for Review

ABC

A cheese vendor currently has 175 pounds of mozzarella available for sale. If each pound of mozzarella sells for \$8.75, which of the following functions gives the amount of mozzarella M , in pounds, still available for sale after d dollars worth has been sold?

A. $M(d) = 175 - \frac{d}{8.75}$

B. $M(d) = 175 - 8.75d$

C. $M(d) = 175 - \frac{8.75}{d}$

D. $M(d) = 175(8.75) - d$

11

★ Mark for Review

ABC

Marcia baked 240 cookies and distributed a third of them evenly among c classmates. Each classmate received 4 cookies. Which equation represents this situation?

A. $\frac{3c}{240} = 4$

B. $\frac{240c}{3} = 4$

C. $\frac{240}{3c} = 4$

D. $\frac{240(3)}{c} = 4$

12

★ Mark for Review

ABC

A taxicab charges \$4.75 for the first mile traveled and \$3.50 for each additional mile. A particular passenger travels x miles in the taxicab and is charged a total of \$24. Which equation represents this situation?

A. $3.50x - 4.75 = 24$

B. $8.25x - 3.50 = 24$

C. $3.50x + 1.25 = 24$

D. $3.50x + 4.75 = 24$

13

★ Mark for Review

ABC

A flask contains 24 milliliters of a 30% acid solution. How many milliliters of an 8% acid solution must be added to the flask to produce a 16% acid solution?

14

★ Mark for Review

ABC

When ordering a certain book in bulk, a school pays \$1,720 for the first 80 copies and a discounted price for each additional copy. The total cost for 200 copies of the book is \$3,700. Which function f gives the total cost, in dollars, for x copies of the book, where $x \geq 80$?

A. $f(x) = 18.50x + 240$

B. $f(x) = 18.50x + 1,720$

C. $f(x) = 16.50x + 400$

D. $f(x) = 16.50x + 1,720$

15

★ Mark for Review

ABC

A tub can be filled by a hose and emptied by a drain. When the drain is closed, the hose can fill $\frac{1}{3}$ of the tub in 1 minute. When the hose is turned off, the drain can empty $\frac{2}{11}$ of the tub in 1 minute. If it takes t minutes for the hose to fill the tub while the drain is open, which of the following equations represents this situation?

A. $\frac{1}{3}t - \frac{2}{11} = 1$

B. $\frac{1}{3} - \frac{2}{11} = \frac{1}{t}$

C. $\frac{1}{3}t + \frac{2}{11} = 1$

D. $\frac{1}{3} + \frac{2}{11} = \frac{1}{t}$

16

★ Mark for Review

ABC

A manufacturer uses two assembly lines to produce refrigerator units. The table summarizes the results of a quality control test used to check for defective units. If a refrigerator unit that was checked during this test is selected at random, what is the probability that the unit was produced by assembly line A, given that the unit is defective?

	Defective	Not defective	Total
Assembly line A	300	5700	6000
Assembly line B	500	3500	4000
Total	800	9200	10000

A. $\frac{1}{20}$

B. $\frac{3}{8}$

C. $\frac{3}{5}$

D. $\frac{5}{8}$

17

★ Mark for Review

ABC

The table shows the results of a survey in which students at three colleges were asked whether they live on campus or off campus. If one of the students surveyed is selected at random, which of the following probabilities has the greatest value?

	Off campus	On campus	Total
College A	335	689	1024
College B	124	482	606
College C	596	274	870
Total	1055	1445	2500

A. The probability that the student lives off campus, given that the student attends College C.

B. The probability that the student lives on campus, given that the student attends College C.

C. The probability that the student attends College C, given that the student lives off campus.

D. The probability that the student attends College C, given that the student lives on campus.

18

★ Mark for Review

ABC

A researcher wanted to determine whether studying at a library is more effective than studying at home. He selected a random sample of 200 students and gave them three days to study for the same test. The 100 students who lived closest to a library were asked to study at a library, and the remaining 100 students were asked to study at home. After the students took the test, the scores of those who studied at a library were compared to the scores of those who studied at home. How should this experiment be changed so that the researcher can conclude whether studying at a library is more effective than studying at home?

- A. The students assigned to study at a library should study at the same library.
- B. The students should be randomly assigned to study either at a library or at home.
- C. The students assigned to study at a library should take a more difficult test than the students assigned to study at home.
- D. No changes to the experiment are necessary.

19

★ Mark for Review

ABC

A group of school administrators wanted to gather opinions on new teacher certification requirements that were introduced in a certain state. The group surveyed a random sample of math teachers from one of the school districts in the state. The survey showed that the majority of the teachers in the sample support the new requirements. Which of the following is the largest population to which the results of the survey can be generalized?

- A. All math teachers in the school district.
- B. All teachers in the school district.
- C. All math teachers in the state.
- D. The teachers in the sample.

20

★ Mark for Review

ABC

All the residents of a town were mailed a survey about proposed renovations to the town's high school. Approximately 10% of the residents completed the survey and mailed it back. Based on the responses, it was estimated that 72% of the residents support the proposed renovations, with an associated margin of error of 12%. Which of the following best explains why these survey results are unlikely to represent the opinions of all the town's residents?

- A. The survey included residents who do not attend the high school.
- B. Not all residents were aware of the proposed renovations before the survey.
- C. Responses to the survey were not obtained from a random sample of the town's residents.
- D. The margin of error is greater than the percentage of residents who responded to the survey.

The tables give the distribution of daily travel times between two towns for bus A and bus B over a 40-day period.

Travel time (minutes)	Frequency
Bus A	
44	5
45	10
47	15
48	10
Bus B	
25	5
30	10
35	15
40	10

Which statement correctly compares the standard deviations of travel times for bus A and bus B?

A. The standard deviation of travel times for bus A is less than the standard deviation of travel times for bus B.

B. The standard deviation of travel times for bus B is less than the standard deviation of travel times for bus A.

C. The standard deviation of travel times for bus A is equal to the standard deviation of travel times for bus B.

D. There is not enough information to compare these standard deviations.

22

★ Mark for Review

ABC

The employees at a new bookstore must stock a certain number of shelves so that the store is ready for its opening in two weeks. The employees stock shelves at a constant rate throughout the two weeks. If $p(t)$ is the number of shelves left to be stocked after t days, which of the following statements best describes the function p ?

A. The function p is an increasing exponential function.

B. The function p is a decreasing exponential function.

C. The function p is an increasing linear function.

D. The function p is a decreasing linear function.

SAT Unit 3

Problem Solving and Data Analysis

Complete Step-by-Step Answer Key

Novel Prep Learning Center

Quick Answer Key

1	650	2	44,400	3	33.75
4	0.46	5	<i>B</i>	6	<i>D</i>
7	<i>B</i>	8	<i>B</i>	9	<i>A</i>
10	<i>A</i>	11	<i>C</i>	12	<i>C</i>
13	42	14	<i>C</i>	15	<i>B</i>
16	<i>B</i>	17	<i>A</i>	18	<i>B</i>
19	<i>A</i>	20	<i>C</i>	21	<i>A</i>
22	<i>D</i>				

Question 1

The number p is 60% of q , so

$$p = 0.60q$$

The number p is 390% of r , so

$$p = 3.90r$$

Set the two expressions equal:

$$0.60q = 3.90r$$

Solve for q :

$$q = \frac{3.90}{0.60}r$$

$$q = 6.5r$$

This means q is 650% of r .

650

Question 2

The starting price is 60% less than \$30,000, so the starting price is 40% of \$30,000.

$$0.40(30,000) = 12,000$$

Eli offers 370% of the starting price:

$$3.70(12,000) = 44,400$$

$$\boxed{44,400}$$

Question 3

The number r is 70% less than s , so

$$r = 0.30s$$

Given

$$s = 180$$

$$r = 0.30(180) = 54$$

The number r is 60% greater than t , so

$$r = 1.60t$$

$$54 = 1.60t$$

$$t = \frac{54}{1.60} = 33.75$$

$$\boxed{33.75}$$

Question 4

The number x is 75% less than y , so

$$x = 0.25y$$

The number z is 84% greater than x , so

$$z = 1.84x$$

Substitute $x = 0.25y$:

$$z = 1.84(0.25y)$$

$$z = 0.46y$$

Therefore, z is 0.46 times y .

$$\boxed{0.46}$$

Question 5

Town A was $n\%$ greater than Town B.

Let Town B be B . Then Town A is

$$B \left(1 + \frac{n}{100} \right)$$

Given Town A is 700:

$$700 = B \left(\frac{100 + n}{100} \right)$$

Solve for B :

$$B = 700 \left(\frac{100}{100 + n} \right)$$

$$\boxed{B}$$

Question 6

Use

$$1 \text{ kilometer} = 0.62 \text{ mile}$$

So

$$1 \text{ mile} = \frac{1}{0.62} \text{ kilometer}$$

For area:

$$1 \text{ square mile} = \left(\frac{1}{0.62} \right)^2 \text{ square kilometers}$$

Thus,

$$5.5 \text{ square miles} = 5.5 \left(\frac{1}{0.62} \right)^2$$

$$= 5.5(2.601\dots)$$

$$\approx 14.3$$

D

Question 7

If a apples cost d dollars, then the cost of 1 apple is

$$\frac{d}{a}$$

The cost of 20 apples is

$$20 \cdot \frac{d}{a}$$

$\frac{20d}{a}$

B

Question 8

The scale is

5 inches : 16 feet

A real width of $10w$ feet corresponds to

$$10w \cdot \frac{5}{16}$$

$$= \frac{50w}{16}$$

$$= \frac{25}{8}w$$

B

Question 9

The rate is

3 square inches per hour

Since

144 square inches = 1 square foot

3 square inches = $\frac{3}{144}$ square feet

Also,

1 foot = 0.3048 meter

So

1 square foot = $(0.3048)^2$ square meters

Thus,

$$\frac{3}{144}(0.3048)^2$$

$$\approx 0.001935$$

Closest answer:

A

Question 10

Each pound costs \$8.75.

If d dollars worth has been sold, then the number of pounds sold is

$$\frac{d}{8.75}$$

The vendor started with 175 pounds, so the amount remaining is

$$M(d) = 175 - \frac{d}{8.75}$$

A

Question 11

Marcia baked 240 cookies.

A third of them is

$$\frac{240}{3}$$

These are distributed evenly among c classmates, so each classmate gets

$$\frac{240}{3c}$$

Each classmate received 4 cookies:

$$\frac{240}{3c} = 4$$

☐ C

Question 12

The first mile costs \$4.75.

Each additional mile costs \$3.50.

If the trip is x miles, then the number of additional miles is

$$x - 1$$

Total cost:

$$4.75 + 3.50(x - 1)$$

Simplify:

$$4.75 + 3.50x - 3.50$$

$$3.50x + 1.25$$

The total charge is \$24, so

$$3.50x + 1.25 = 24$$

☐ C

Question 13

Let x be the number of milliliters of 8% acid solution added.

Initial acid amount:

$$0.30(24) = 7.2$$

Acid added:

$$0.08x$$

Total solution amount:

$$24 + x$$

Final solution is 16%, so

$$7.2 + 0.08x = 0.16(24 + x)$$

Expand:

$$7.2 + 0.08x = 3.84 + 0.16x$$

Subtract $0.08x$:

$$7.2 = 3.84 + 0.08x$$

Subtract 3.84:

$$3.36 = 0.08x$$

$$x = \frac{3.36}{0.08} = 42$$

$$\boxed{42}$$

Question 14

The first 80 copies cost \$1,720.

The total cost for 200 copies is \$3,700.

So the cost of the additional 120 copies is

$$3700 - 1720 = 1980$$

Discounted price per additional copy:

$$\frac{1980}{120} = 16.50$$

For $x \geq 80$, the total cost is

$$f(x) = 1720 + 16.50(x - 80)$$

Simplify:

$$f(x) = 1720 + 16.50x - 1320$$

$$f(x) = 16.50x + 400$$

C

Question 15

The hose fills

$$\frac{1}{3}$$

of the tub per minute.

The drain empties

$$\frac{2}{11}$$

of the tub per minute.

So the net filling rate is

$$\frac{1}{3} - \frac{2}{11}$$

If it takes t minutes to fill the tub, then the net rate is

$$\frac{1}{t}$$

Therefore,

$$\frac{1}{3} - \frac{2}{11} = \frac{1}{t}$$

B

Question 16

We need

$$P(\text{Assembly line A} \mid \text{Defective})$$

This means:

$$\frac{\text{Defective from A}}{\text{Total defective}}$$

From the table:

$$\text{Defective from A} = 300$$

$$\text{Total defective} = 800$$

So

$$\begin{aligned} P &= \frac{300}{800} \\ &= \frac{3}{8} \end{aligned}$$

B

Question 17

Compute each probability.

Choice A:

$$P(\text{Off campus} \mid \text{College C}) = \frac{596}{870} \approx 0.685$$

Choice B:

$$P(\text{On campus} \mid \text{College C}) = \frac{274}{870} \approx 0.315$$

Choice C:

$$P(\text{College C} \mid \text{Off campus}) = \frac{596}{1055} \approx 0.565$$

Choice D:

$$P(\text{College C} \mid \text{On campus}) = \frac{274}{1445} \approx 0.190$$

The greatest value is choice A.

A

Question 18

The issue is that the researcher assigned students based on how close they lived to a library. That creates a possible confounding variable.

To determine whether studying at a library is more effective than studying at home, the students should be randomly assigned to the two study locations.

B

Question 19

The sample was a random sample of math teachers from one school district.

Therefore, the results can be generalized to the population from which the random sample was taken:

all math teachers in that school district

It cannot be generalized to all teachers or all math teachers in the state.

A

Question 20

Only about 10% of residents responded.

The responses were not obtained from a random sample; instead, they came from people who chose to respond.

This is a voluntary response sample, which can be biased.

C

Question 21

Standard deviation measures spread.

Bus A travel times:

44, 45, 47, 48

Bus B travel times:

25, 30, 35, 40

Bus B's times are much more spread out than Bus A's times.

Therefore,

standard deviation of Bus A < standard deviation of Bus B

A

Question 22

The employees stock shelves at a constant rate.

A constant rate means a linear function.

Since $p(t)$ is the number of shelves left to be stocked, this number decreases as time passes.

Therefore, $p(t)$ is a decreasing linear function.

D